Learner's Workbook

Worksheets and exercises for the 40-hour CDF intensive

SpaceCDF -- AI-Assisted Concurrent Design Facility
University of Ottawa - SEDTI
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DRAFT

Learner's Workbook

How to use this workbook

This workbook is your hands-on companion through the 40-hour SpaceCDF intensive. There is one worksheet per session. Each worksheet contains:

- Key Equations Reference -- the essential formulas you need for that session
- Calculation Space -- room to work through problems step by step
- Budget Templates -- tables to fill in as you design your mission
- Notes & Reflections -- space to capture insights and discussion points

The workbook is paired with the Facilitator's Book, which carries the teaching content, worked solutions, and references.

> *Tip: Complete the expected reading listed at the top of each worksheet before the session -- the exercises are designed to be completed quickly when you arrive prepared.*

Worksheet 1.1: Introduction to Space Mission Design

Name:	____
Date:	____
Team:	____
Session:	1.1 -- Introduction to Space Mission Design

Part A: The 17 SE Processes -- SpaceCDF Mapping

For each NASA SE process (NPR 7123.1D), identify which SpaceCDF feature implements it. Write the tab name, button, or panel. If the process requires human judgment that the tool cannot automate, mark "Manual" in the last column.

Hint: Navigate through SpaceCDF's workflow steps (Need -> Concept -> Requirements -> Design) and explore each tab. Many processes map to more than one feature.

#	SE Process	SpaceCDF Feature(s)	Manual?
1	Stakeholder Expectations Definit	______________________	_____
2	Technical Requirements Definitic	______________________	_____
3	Logical Decomposition	______________________	_____
4	Design Solution Definition	______________________	_____
5	Product Implementation	______________________	_____
6	Product Integration	______________________	_____
7	Product Verification	______________________	_____
8	Product Validation	______________________	_____
9	Product Transition	______________________	_____
10	Technical Planning	______________________	_____
11	Requirements Management	______________________	_____
12	Interface Management	______________________	_____
13	Technical Risk Management	______________________	_____
14	Configuration Management	______________________	_____
15	Technical Data Management	______________________	_____
16	Technical Assessment	______________________	_____
17	Decision Analysis	______________________	_____

Discussion question: Which processes are well-supported by the tool? Which ones fundamentally require human judgment and cannot be fully automated? Write your answer below:

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Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase (Pre-A through F) it belongs to and which review gate marks the end of that phase:

Activity	NASA Phase	Review Gate
"We need to monitor sea ice thickness in the	________	________
Writing system-level "shall" requirements wi	________	________
Selecting a specific reaction wheel vendor a	________	________
Running thermal-vacuum qualification tests	________	________
Operating the satellite and producing daily d	________	________
Decommissioning the satellite and performir	________	________
Preliminary orbit trade study comparing SSC	________	________

Detailed finite element analysis of the prima	__________	__________

Hint: Refer to the lifecycle phase table in Session 1.1, Section 4.3. Ask yourself: "What level of design detail does this activity require?"

Part C: CDF Position Interaction Analysis

For each design conflict below, identify which engineering positions must interact to resolve it. Then briefly describe how you would approach the resolution.

Conflict 1: The payload requires 50 W but the solar array can only provide 30 W.

Positions involved: ____

Resolution approach: ____

____

Conflict 2: The reaction wheel vibration exceeds the payload's pointing stability requirement.

Positions involved: ____

Resolution approach: ____

____

Conflict 3: The selected orbit (700 km) requires a propulsive deorbit for debris compliance, but there is no mass budget remaining for a propulsion system.

Positions involved: ____

Resolution approach: ____

____

Conflict 4: The X-band downlink antenna pattern interferes with the payload's sensitive optical receiver.

Positions involved: ____

Resolution approach: ____

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Part D: SpaceCDF Tool Exploration

Navigate through the SpaceCDF tool and answer each question:

1. How many KPI cards are shown on the Design Dashboard? List them by name:

____

____

____

2. In the Gate Review tab, what are the MCR exit criteria? Which are auto-evaluated?

____

____

____

3. What engineering positions are available when starting a collaborative session?

____

____

4. What workflow steps does SpaceCDF guide you through (from left to right)?

____

5. In the Positions tab, find your assigned position. What key questions does it list?

____

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____

Part E: V-Model Sketch

In the space below, sketch the System-V model from memory. Label each level on both the left side (decomposition) and right side (integration). Draw the horizontal traceability arrows and label them.

Use this space for your sketch:

Notes & Reflections

Record key points from the group discussion, questions you want to follow up on, and connections to your mission concept:

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Worksheet 1.2: Canadian Space Ecosystem & Regulatory Environment

Name:	______________________________
Date:	______________________________
Team:	______________________________
Session:	1.2 -- Canadian Space Ecosystem & Regulatory Environment

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Part A: Canadian Space Industry Mapping

For your team's mission concept, identify potential Canadian industry partners at each tier. If no Canadian supplier exists for a capability, note what country you would source from and why.

Tier	Capability Needed	Canadian Candidate(s)	International Alternative	Justification
Tier 1 (Prime/Integration)	_______________	_______________	_______________	_______________
Tier 2 (Subsystem)	_______________	_______________	_______________	_______________
Tier 2 (Subsystem)	_______________	_______________	_______________	_______________
Tier 3 (Component/Servic	_______________	_______________	_______________	_______________
Tier 4 (R&D/Academic)	_______________	_______________	_______________	_______________

Question: What CSA funding program(s) might support your mission? Why?

____

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Part B: RSSSA Compliance Assessment

Answer the following questions for your team's mission concept:

1. Does your mission involve any remote sensing of Earth (imaging, SAR, radiometry, etc.)?

____

2. If yes, will the satellite be operated from Canada?

____

3. What RSSSA conditions might apply? (Check all that apply)

Condition	Applies?	Notes
Priority access for Government of Canada	________	________________
Shutter control provisions	________	________________
Data disposition / handling plan	________	________________
System disposal plan	________	________________
Raw data protection	________	________________
Interoperability arrangements (foreign acces	________	________________

4. If your mission does NOT involve remote sensing, what other regulatory requirements apply? (e.g., spectrum licensing, export control)

____

____

Part C: Spectrum Licensing Plan

Identify the frequency bands your mission will use and the licensing path for each:

Link	Band	Frequency (GHz)	Data Rate	Use (TT&C / Payloa	Licensing Path
Uplink	________	________	________	________	________

Downlink (TT&C)	__________	__________	__________	__________	__________
Downlink (Payload)	__________	__________	__________	__________	__________
Other	__________	__________	__________	__________	__________

Hint: For a typical CubeSat, TT&C uses S-band (2.0--2.3 GHz) and payload data uses X-band (8.025--8.4 GHz). The licensing path for each is: ISED application -> ISED technical review -> ITU filing (API) -> ITU coordination -> ISED licence issuance.

Question: How early in your project should you begin the spectrum licensing process? Why?

____

____

____

Part D: Export Control Classification

List the 3 most critical components in your preliminary mission concept. For each, identify the likely export control classification:

Component	Country of Origin	Likely Classification	Licence Required?
_____________________	__________	ECCN: __________ / USML Cat	__________
_____________________	__________	ECCN: __________ / USML Cat	__________
_____________________	__________	ECCN: __________ / USML Cat	__________

Hint: Most commercial-off-the-shelf (COTS) space components from US manufacturers are classified under EAR (ECCN 9A515 for spacecraft parts). Military-grade or specifically designed items may fall under ITAR (USML Category XV). Canadian-origin components are subject to EIPA if they appear on the Export Control List.

Question: If your mission includes a US-origin star tracker and you plan to launch on an Indian PSLV, what export control issues arise?

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Part E: Regulatory Timeline

Based on your mission concept, estimate when each regulatory activity should begin and how long it will take. Mark the activities on the timeline below:

Regulatory Activity	Start Phase	Duration	Critical Path?
RSSSA licence application	__________	__________	__________
ISED spectrum application	__________	__________	__________
ITU coordination	__________	__________	__________
Export permit(s)	__________	__________	__________
Launch provider interface agreer	__________	__________	__________

Question: Which regulatory activity is most likely to be on the critical path for your mission? Why?

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Notes & Reflections

Record key points from the discussion, regulatory questions you need to research further, and implications for your mission design:

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Worksheet 1.3: International Standards for Space Systems

Name:	______________________________
Date:	______________________________
Team:	______________________________
Session:	1.3 -- International Standards for Space Systems

Part A: Standard Applicability Matrix

For your team's mission concept, determine which ECSS and NASA standards apply. For each standard, decide whether it is fully applicable, partially applicable (with tailoring), or not applicable. Justify your decision.

Standard	Branch	Applicability	Tailoring Notes / Justification
ECSS-M-ST-10C (Project Manag	M	__________	______________________________
ECSS-M-ST-40C (Configuration	M	__________	______________________________
ECSS-M-ST-80C (Risk Manager	M	__________	______________________________
ECSS-E-ST-10C (Systems Engi	E	__________	______________________________
ECSS-E-ST-20C (Electrical & El	E	__________	______________________________
ECSS-E-ST-32C (Structures)	E	__________	______________________________
ECSS-E-ST-40C (Software)	E	__________	______________________________
ECSS-E-ST-50C (Communicatio	E	__________	______________________________
ECSS-Q-ST-10C (Product Assur	Q	__________	______________________________
ECSS-U-AS-10C (Debris Mitigat	U	__________	______________________________
ISO 24113 (Debris Mitigation)	--	__________	______________________________
NPR 7123.1D (SE Processes)	--	__________	______________________________

Hint: A university CubeSat mission would typically invoke ECSS-E-ST-10C with heavy tailoring, ECSS-U-AS-10C fully, and might waive ECSS-Q-ST-10C (product assurance) due to cost constraints. A CSA-funded mission would typically require more complete ECSS compliance.

Question: What is the minimum set of standards your mission MUST comply with regardless of mission class?

____

____

____

Part B: CDS Rev 14 Compliance Checklist

If your mission uses a CubeSat form factor, complete this compliance checklist. If not, skip to Part C.

CDS Requirement	Your Design Value	Compliant?	Notes
Unit dimensions (100 x 100 x 11	__________	__________	__________
Mass per U ($\leq$ 2.0 kg)	__________	__________	__________
Total mass (for your form factor)	__________	__________	__________
Rail material (hard-anodised Al)	__________	__________	__________
CG within 2 cm of geometric cen	__________	__________	__________
No protrusions beyond envelope	__________	__________	__________
RF silence for 30 min after deplo	__________	__________	__________
Deployment switches on all depl	__________	__________	__________
Propulsion inhibits (3 independe	__________	__________	__________

Hint: Use SpaceCDF's Compliance panel to check CDS compliance. The tool will flag non-compliant parameters automatically based on your mass and dimensions entries.

Part C: Debris Mitigation Compliance

Complete the following debris mitigation assessment for your mission orbit:

1. Planned orbit altitude: __________ km
2. Estimated natural orbital lifetime (without propulsion):

Use the rule of thumb: $\tau_{years} \approx \frac{h - 200}{30} \times \frac{m/A}{50}$ where h is altitude (km), m is mass (kg), A is cross-section area (m^2)

h = __________ km, m = __________ kg, A = __________ m^2

m/A = __________ kg/m^2

τ = __________ years (approximate)

3. Compliance assessment:

Requirement	Limit	Your Mission	Compliant?
IADC post-mission lifetime	$\leq$ 25 years	__________	__________
FCC post-mission lifetime (2024-	$\leq$ 5 years	__________	__________
ISO 24113 disposal reliability	$\geq$ 90%	__________	__________

4. If non-compliant with the 5-year FCC rule, what options exist?

____

____

____

5. Passivation plan (list all stored energy sources and how you will deplete them at end of life):

Energy Source	Passivation Method
Battery	________________________________
Pressure vessels (if any)	________________________________
Reaction wheels	________________________________
RF transmitter	________________________________
Other: __________	________________________________

Part D: Structural Design Margins

If your mission has preliminary structural design information, calculate the margin of safety:

Use the formula: $MoS = \frac{\sigma_{allowable}}{FoS \times \sigma_{applied}} - 1$

Worked example: For Al 7075-T6 (σ_{yield} = 503 MPa), applied stress = 200 MPa, FoS (yield) = 1.25:

$MoS = \frac{503}{1.25 \times 200} - 1 = \frac{503}{250} - 1 = 1.012$

This is a positive margin (compliant).

Your calculation:

Material: ___________ , $\sigma_{allowable}$ = ___________ MPa

$\sigma_{applied}$ = ___________ MPa, FoS = ___________

MoS = ___________ (show work below)

____

____

Part E: SpaceCDF Compliance Panel

Navigate to the Compliance panel in SpaceCDF and answer:

1. What is the debris mitigation compliance status for your mission?

____

2. What orbital altitude would be needed to comply with the FCC 5-year rule WITHOUT propulsion? (Use the tool's orbital lifetime calculator)

____

3. What standards has the tool automatically suggested for your mission type?

____

____

4. Are there any compliance flags (warnings or errors) on your current design?

____

____

Notes & Reflections

Record key points about standards applicability, compliance challenges, and design implications for your mission:

____

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Worksheet 1.4: Mission Needs, Trade Studies & Concept of Operations

Name:	______________________________
Date:	______________________________
Team:	______________________________
Session:	1.4 -- Mission Needs, Trade Studies & ConOps

Part A: Stakeholder Analysis Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, priority, and influence level.

#	Stakeholder	Role	Key Needs	Constraints Impo	Priority (P/S/T)	Influence (H/M/L)
1	____________	____________	______________	______________	______	______
2	____________	____________	______________	______________	______	______
3	____________	____________	______________	______________	______	______
4	____________	____________	______________	______________	______	______
5	____________	____________	______________	______________	______	______
6	____________	____________	______________	______________	______	______

P = Primary (must satisfy), S = Secondary (should satisfy), T = Tertiary (nice to have)

H = High influence, M = Medium, L = Low

Question: Which stakeholder has the most influence over your mission? Do any stakeholders have conflicting needs? If so, how might you resolve the conflict?

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Part B: Problem Statement

Write your team's mission problem statement. It must answer all four questions: What is the problem? Who is affected? What is the impact? What constraints exist?

Problem Statement:

____

____

____

____

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____

Peer Review Checklist (exchange with another team):

Criterion	Pass?	Feedback
Describes WHAT is needed, not HOW to ac	______	__________________________________
Identifies the capability gap?	______	__________________________________
Names specific affected stakeholders?	______	__________________________________
Quantifies the impact of not solving it?	______	__________________________________
States constraints (budget, schedule, regula	______	__________________________________
Free of solution language? (No specific sate	______	__________________________________

Part C: Mission Objectives

Write 2--3 mission objectives. Score each against the quality checklist.

Objective 1: ____

____

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?	______	__________________________________
Has measurable criterion with number + unit	______	__________________________________
Achievable with current technology (TRL \ge	______	__________________________________
Relevant to the problem statement?	______	__________________________________
Traceable to a specific stakeholder need?	______	__________________________________
Classified as primary or secondary?	______	__________________________________

Objective 2: ____

____

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?	______	__________________________________
Has measurable criterion with number + unit	______	__________________________________
Achievable with current technology (TRL \ge	______	__________________________________
Relevant to the problem statement?	______	__________________________________
Traceable to a specific stakeholder need?	______	__________________________________
Classified as primary or secondary?	______	__________________________________

Objective 3 (optional): ____

____

Part D: MoE / MoP / TPM Trace

For your primary objective, trace the measurement hierarchy:

Objective: ____

MoE (Measure of Effectiveness -- operational effectiveness from user perspective):

____

____

MoP (Measure of Performance -- technical performance of the system):

____

____

TPM (Technical Performance Measure -- design parameter tracked during development):

____

____

Hint: If your objective is "Provide 10 m GSD multispectral imagery...", then:

- MoE might be "% of target area with usable (<30% cloud) imagery within 5 days"
- MoP might be "GSD \leq 10 m at nadir, SNR \geq 100 in all bands"
- TPM might be "current estimate of aperture diameter (target: 15 cm), imager mass (target: 1.5 kg)"

Part E: Trade Study -- Space vs Non-Space

Complete a trade study evaluating whether a dedicated satellite is the right solution for your mission need.

Decision statement: ____

Alternatives:

#	Alternative	Description
1	______________________	____
2	______________________	____
3	______________________	____
4	Do nothing / status quo	____

Criteria and Weights (must sum to 1.0):

Use pairwise comparison or direct assignment. Show your weight derivation below.

Criterion	Weight	Direction (higher/lower is better)
________________________	_____	_________
________________________	_____	_________
________________________	_____	_________
________________________	_____	_________
________________________	_____	_________
Total	1.00	

Weight derivation (pairwise comparison table or notes):

____

____

____

Scoring Matrix (normalised 0--1):

Alternative	Criterion 1	Criterion 2	Criterion 3	Criterion 4	Criterion 5	Weighted Total
1: ___________	_____	_____	_____	_____	_____	_____
2: ___________	_____	_____	_____	_____	_____	_____
3: ___________	_____	_____	_____	_____	_____	_____
4: Do nothing	_____	_____	_____	_____	_____	_____

Recall: Weighted Total = \sum w_c \times s(a,c)

Result: The highest-scoring alternative is: ____

Decision statement (1 paragraph -- include top 2 alternatives and why one was selected over the other):

____

____

____

____

____

Sensitivity check: If you increase the weight of your second-most-important criterion by 20% (and re-normalise), does the ranking change?

____

____

Part F: Concept of Operations Outline

Document your preliminary ConOps following the NASA SEH Appendix S structure:

1. Mission Architecture:

Space segment: ____

Ground segment: ____

User segment: ____

2. Orbit-Average Power Budget:

Calculate using: $P_{avg} = \sum P_{mode,i} \times DC_i$

Mode	Power (W)	Duty Cycle (fraction of orbit)	P x DC (W)
Idle / Housekeeping	__________	__________	__________
Science / Imaging	__________	__________	__________
Downlink	__________	__________	__________
Eclipse	__________	__________	__________
		Orbit-average power:	__________ W

Solar array requirement: P_{SA} = __________ W

Show your calculation:

____

____

3. Data Budget:

Daily data generation: __________ GB/day

Downlink capacity per pass: __________ GB/pass

Passes per day: __________

Daily downlink capacity: __________ GB/day

Balance (downlink $\geq$ generation)? __________

Part G: SpaceCDF Tool Entry

In SpaceCDF, enter your mission need (Step 1) and concept (Step 2). Then answer:

1. In Step 1, did the tool auto-detect your mission type from the objective text? What type did it suggest?

2. In Step 2, run the mission trade analysis. Does the tool's recommendation match your manual trade study result from Part E?

3. In the ConOps tab, build your architecture diagram. How many systems did you define?

4. In the Trade Studies tab, run an orbit selection trade with at least 3 orbit options. What orbit did it recommend?

Notes & Reflections

Record key decisions made today, open questions, and what you want to investigate further:

Worksheet 2.1: The System-V and Requirements Engineering

Name:	________________________________
Date:	________________________________
Team / Mission:	________________________________
Role / Position:	________________________________

SpaceCDF Tab: Requirements

Key Equations Reference

> *Traceability Coverage:*
>
> *$\text{Coverage} = \frac{N_{\text{verified requirements}}}{N_{\text{total requirements}}} \times 100\%$*
>
> *Target: 100% -- every requirement needs a parent trace, a verification method, and a responsible subsystem.*

> *Requirement Quality (SMART):*
>
> *Specific -- Measurable -- Achievable -- Relevant -- Traceable*

> *Verification Methods (ATID):*
>
> *Analysis -- Test -- Inspection -- Demonstration*

Part A: System-V Mapping (10 min)

For your mission, identify one element at each level of the V-model:

V-Level	Left Branch (Decomposition)	Right Branch (Verification)
Stakeholder Need		
Mission Requirement		
System Requirement		
Subsystem Requirement		

____

____

____

Part B: Requirement Quality Assessment (15 min)

Evaluate each requirement using the SMART checklist and NASA Appendix C criteria. Mark Pass (P) or Fail (F). Rewrite any failures.

#	Requirement	S	M	A	R	T	WHAT or HO	Rewrite (if ne
1	"The spacecra							
2	"The system s							
3	"The EPS sha							
4	"The system s							
5	"The structure							

____

____

____

____

Part C: Requirement Hierarchy and Traceability (15 min)

For your mission, write one requirement at each level. Draw the traceability chain.

Stakeholder Objective (traces FROM): ____

____

Mission Requirement (MR-___): ____

____

System Requirement (SR-___), derived from MR above: ____________________________

____

Subsystem Requirement (SSR-___), derived from SR above: _______________________

____

Traceability diagram (draw arrows showing parent-child relationships):

Part D: Splitting Compound Requirements (10 min)

Split this compound requirement into individually testable statements:

> *"The system shall provide 10 m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."*

#	Split Requirement (one	Type (F/P/I/C/E)	Verification (A/T/I/D)	Phase
1				
2				
3				
4				
5				

Part E: SpaceCDF Requirements Generation (20 min)

1. In SpaceCDF, click "Generate from Objectives." How many requirements were generated? ____

2. Use the Level filter -- count at each level:
- Mission: ____   System: ____   Subsystem: ____

3. Classify 5 requirements by type:

4. Find one requirement that says HOW (implementation) not WHAT. Write it and your correction:

- Original: ____
- Corrected: ____

5. Accept at least 5 requirements and reject at least 1. Record your decisions:

6. For 5 key requirements, assign verification method and phase:

Notes & Reflections

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Worksheet 2.2: Functional Decomposition and Allocation

Req ID	Requirement Text (abbreviated	Type (Functional / Performanc	
Req ID	Action (Accept/Edit/Reject)	Rationale	
Req ID	Verification Method (A/T/I/D)	Phase (B/C/D/E)	Rationale for Method
Name:	____________________		
Date:	____________________		
Team / Mission:	____________________		
Role / Position:	____________________		

SpaceCDF Tab: Functions

Key Equations Reference

> *Function Types: Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support*
>
> *Universal Functions (every spacecraft):*
>
> *1. Generate electrical power*
>
> *2. Maintain thermal environment*
>
> *3. Survive launch environment*
>
> *4. Communicate with ground (TTC)*
>
> *5. Dispose at end of life*
>
> *Coverage Rule: Every leaf function must trace to at least one requirement. No gaps.*
>
> *Allocation Rule: Multi-allocated functions create interfaces that must be explicitly managed.*

Part A: Mission Function Tree (20 min)

Draw your mission's complete function tree. Use minimum 3 levels of decomposition. Include both mission-specific and universal functions.

Mission-specific functions:

```
F-001: _______________________________________________
  +-- F-002: _______________________________________________
      +-- F-002a: ___________________________________________
      +-- F-002b: ___________________________________________
  +-- F-003: _______________________________________________
      +-- F-003a: ___________________________________________
  +-- F-004: _______________________________________________
  +-- F-005: _______________________________________________
```

Universal functions:

```
F-010: Generate electrical power
  +-- F-011: _______________________________________________
  +-- F-012: _______________________________________________
F-020: Maintain thermal environment
  +-- F-021: _______________________________________________
F-030: Survive launch environment
F-040: Communicate with ground (TTC)
F-050: Dispose at end of life
```

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Part B: Function-to-Requirement Traceability (15 min)

For each leaf function, write one derived requirement with a measurable threshold:

____

____

Part C: Multi-Allocation Analysis (10 min)

Identify at least one function that is allocated to more than one subsystem:

Function: ____

Subsystem A (primary): ____

Subsystem B (contributor): ____

Interface created between A and B: ____

____

Who "owns" this function? ____

What interface requirements are needed? ____

____

____

Part D: Performance Criteria (10 min)

For 4 key functions, define quantitative performance criteria:

Part E: SpaceCDF Coverage Check (15 min)

After entering functions in SpaceCDF:

1. How many total functions in your tree? ____
2. How many leaf functions? ____
3. How many leaf functions have NO linked requirements (amber badge)? ____
4. List the uncovered functions and derive requirements for them:
5. Are there any functions the tool generated that are NOT appropriate for your mission? List and explain why you removed them:

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Notes & Reflections

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Worksheet 2.3: Orbit Selection and Mission Architecture

Function ID	Function Name	Allocated To (Subsyster	Derived Requirement	Type
Function	Performance Criterion	Value	Unit	Source
Function ID	Function Name	Derived Requirement		
Name:	____________________			
Date:	____________________			
Team / Mission:	____________________			
Role / Position:	____________________			

SpaceCDF Tabs: Mission Architecture, Orbit Trade Advisor

Key Equations Reference

> *Orbital period:* $T = 2\pi \sqrt{a^3/\mu}$ *where* $a = R_E + h$, $\mu = 3.986 \times 10^{14}$ *m^3/s^2,* $R_E = 6371$ *km*
>
> *Orbital velocity:* $v = \sqrt{\mu/a}$
>
> *Eclipse fraction:* $f_{\text{ecl}} = \frac{1}{\pi} \arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right)$
>
> *Sun-synchronous inclination:* $\cos(i) = -\frac{2\dot{\Omega} a^{7/2}}{3 R_E^2 J_2 \sqrt{\mu}}$ *where* $J_2 = 1.0826 \times 10^{-3}$
>
> *Hohmann* ΔV*:* $\Delta V_1 = \sqrt{\mu/r_1}\left(\sqrt{2r_2/(r_1+r_2)} - 1\right)$
>
> *Swath width:* $W = 2h\tan(\theta)$

Part A: Orbital Mechanics Calculations (20 min)

Your selected orbit: Altitude h = _____ km,   Inclination i = _____ deg,   Type: __________

Show all working for each calculation:

1. Semi-major axis:

a = R_E + h = 6371 + _____ = _____ km = _____ m

____

2. Orbital period:

T = 2\pi\sqrt{a^3/\mu} = 2\pi\sqrt{ _____ ^3 /\ 3.986 \times 10^{14}} = _____ s = _____ min

3. Orbital velocity:

v = \sqrt{\mu/a} = \sqrt{3.986 \times 10^{14} /\ } _____ = _____ m/s

4. Eclipse fraction (maximum):

f = \frac{1}{\pi}\arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right) = \frac{1}{\pi}\arccos\left(\frac{\sqrt{\ }}{\ }\right) = _____

5. Sunlight and eclipse time per orbit:

t_{\text{sun}} = T \times (1 - f) = _____ \times _____ = _____ min

t_{\text{ecl}} = T \times f = _____ \times _____ = _____ min

6. Sun-synchronous inclination (if applicable):

\cos(i) = _____ \Rightarrow i = _____ deg

Does this match your selected inclination? Y / N

Part B: Orbit Trade Matrix (10 min)

From SpaceCDF's Orbit Trade Advisor, record the top 3 candidates:

Rank	Orbit Type	Alt (km)	Inc (deg)	GSD (m)	Revisit (d)	Natural Lifeti	FCC 5-yr?	Score
1								
2								
3								

Selected orbit: ___________ Rationale: ___________________________________

____

____

Part C: Debris Compliance (10 min)

For your selected orbit:

Parameter	Value
Natural orbital lifetime	_____ years
FCC 5-year compliant?	Y / N
IADC 25-year compliant?	Y / N
Propulsion needed for deorbit?	Y / N
If yes, estimated deorbit \Delta V	_____ m/s
Deorbit method (if needed)	____

Show \Delta V calculation if propulsion needed:

____

____

____

Part D: Downstream Cascade Analysis (10 min)

How does your orbit selection affect each subsystem? Fill in the impact:

Subsystem	Orbit Parameter That Matters	Impact on Your Design
Payload (GSD)	Altitude = _____ km	GSD = _____ m
EPS (eclipse)	Eclipse fraction = _____	Eclipse duration = _____ min
Comms (link)	Slant range at 10 deg elev = _____ km	FSPL increase = _____ dB
Thermal	Eclipse/sunlight cycling	Hot case / Cold case concern: ______
AOCS	Orbit rate = _____ deg/min	Nadir tracking rate needed
Propulsion	Lifetime = _____ yr	Propulsion needed? Y / N
Radiation	TID estimate = _____ krad/yr	Electronics class: _____________

Part E: Discussion (10 min)

What would happen to your mission if you moved the orbit 100 km higher?

___

___

___

What would happen if you moved it 100 km lower?

___

___

___

Notes & Reflections

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Worksheet 2.4: Mission Architecture -- Segments, Interfaces, and Budgets

Name:	___
Date:	___

Team / Mission:	____
Role / Position:	____

SpaceCDF Tabs: System Architecture, Interfaces, Dashboard, Budget Breakdown

---

Key Equations Reference

> *Mass margin: \text{Margin}_\% = \frac{M_{\text{alloc}} - M_{\text{MEV}}}{M_{\text{alloc}}} \times 100\%*
>
> *   Green: > 20% | Amber: 10--20% | Red: < 10%*
>
> *MEV: M_{\text{MEV}} = M_{\text{CBE}} + \text{equipment margin} + \text{system margin}*
>
> *Orbit-average power: P_{\text{avg}} = \sum (P_{\text{mode}} \times f_{\text{duty}})*
>
> *SA power: P_{\text{SA}} = P_{\text{peak}} + \frac{P_{\text{ecl}} \times t_{\text{ecl}}}{t_{\text{sun}} \times \eta_{\text{charge}}}*
>
> *ECSS Phase A margins: Equipment 20% + System 20% = ~44% compound*

Part A: Mass Budget (15 min)

Complete from your SpaceCDF design (or estimate if pre-run):

Subsystem	CBE Mass (kg)	% of Dry	Equip. Margin (20%)	MEV (kg)
Payload				
EPS				
AOCS				
Comms (TTC)				
OBC				
Thermal				
Structure				
Harness				
Propulsion				
Dry Total (CBE)		100%		
System margin (20%)				
Dry MEV				
Propellant				

Wet Mass				
Launcher Allocation				
Mass Margin				
Mass Margin %			G / A / R	

Show margin calculation:

____

____

Part B: Power Budget by Mode (15 min)

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC					
AOCS					
Payload					
Comms (TX)					
Thermal					
Total					
Duty cycle (%)					

Orbit-average power calculation (show working):

$P_{\text{avg}} = \underline{\qquad} \times \underline{\qquad} + \underline{\qquad} \times \underline{\qquad} + \underline{\qquad} \times \underline{\qquad} + \underline{\qquad} \times \underline{\qquad} = \underline{\qquad}$ W

____

SA power required:

$P_{\text{recharge}} = \frac{P_{\text{ecl}} \times t_{\text{ecl}}}{t_{\text{sun}} \times \eta_{\text{charge}}} = \frac{\ \ \ \ \times \ \ \ \ }{\ \ \ \ \times 0.9} = \underline{\qquad}$ W

$P_{\text{SA,EOL}} = P_{\text{peak}} + P_{\text{recharge}} = \underline{\qquad} + \underline{\qquad} = \underline{\qquad}$ W

$P_{\text{SA,BOL}} = P_{\text{SA,EOL}} / (1 - 0.025)^n = \underline{\qquad} / \underline{\qquad} = \underline{\qquad}$ W

____

Part C: Interface Identification (15 min)

List the 5 most critical interfaces in your design:

#	Subsystem A	Subsystem B	Interface Types (M/E/T/D	Key Concern
1				
2				
3				
4				
5				

Write 2 formal interface requirements for your most critical pair:

Interface: _____________ <-> _____________

IR-001: ____

____

IR-002: ____

____

Verification method for IR-001: _____   IR-002: _____

Part D: Conflict Resolution (10 min)

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict description: ____

____

Severity: Critical / Major / Minor

Resolution options considered:

#	Option	Pros	Cons
1			
2			
3			

Selected resolution: ____

Rationale: ____

Part E: Budget Health Check (10 min)

From the SpaceCDF Dashboard, record all KPIs:

Budget	Value	Margin	Status (G/A/R)
Mass	_____ kg wet vs _____ kg alloc	_____%	
Power (worst mode)	_____ W vs _____ W SA	_____%	
Link	_____ dB margin	>= 3 dB?	
Cost	_____ MEUR vs _____ ceiling	_____%	
Pointing	_____ deg vs _____ deg req	_____%	

Which budget is tightest? ____

What single design change would improve it? __________________________

____

Notes & Reflections

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____

____

Worksheet 3.1: Power System and Thermal Control Design

Name:	______________________________

Date:	______________________________________
Team / Mission:	______________________________________
Role / Position:	______________________________________

SpaceCDF Tabs: Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

---

Key Equations Reference

> *SA EOL power: $P_{\text{SA,EOL}} = P_{\text{peak}} + \frac{P_{\text{ecl}} \times t_{\text{ecl}}}{t_{\text{sun}} \times \eta_{\text{charge}}}$*
>
> *SA BOL power: $P_{\text{SA,BOL}} = \frac{P_{\text{SA,EOL}}}{(1 - \delta)^n}$ ($\delta = 0.025$/yr for GaAs)*
>
> *SA area: $A = \frac{P_{\text{BOL}}}{\eta_{\text{cell}} \times S \times \cos\theta \times f_{\text{pack}}}$ ($\eta = 0.295$, $S = 1361$ W/m^2, $f_{\text{pack}} = 0.85$)*
>
> *Battery capacity: $C = \frac{P_{\text{ecl}} \times t_{\text{ecl}}}{DOD \times \eta_{\text{discharge}}}$ ($DOD = 0.30$, $\eta = 0.95$)*
>
> *Thermal equilibrium: $T = \left(\frac{Q_{\text{abs}} + Q_{\text{int}}}{\varepsilon \sigma A_{\text{rad}}}\right)^{1/4}$ ($\sigma = 5.67 \times 10^{-8}$ W/m^2/K^4)*
>
> *ECSS thermal margins: Operating +/-5 degC, Acceptance +/-10 degC, Qualification +/-15 degC*

Part A: Solar Array Sizing (20 min)

Show all calculations step by step.

1. Peak sunlight power demand: P_{peak} = _____ W (from mode: ______________)

2. Eclipse power demand: P_{ecl} = _____ W

3. Eclipse and sunlight times (from Worksheet 2.3):

t_{ecl} = _____ min t_{sun} = _____ min

4. Recharge power:

P_{\text{recharge}} = \frac{P_{\text{ecl}} \times t_{\text{ecl}}}{t_{\text{sun}} \times \eta_{\text{charge}}} = \frac{\ \ \ \ \ \times \ \ \ \ \ }{\ \ \ \ \ \times 0.9} = _____ W

5. SA EOL required:

P_{\text{SA,EOL}} = P_{\text{peak}} + P_{\text{recharge}} = _____ + _____ = _____ W

6. EOL degradation factor (mission lifetime = _____ years, \delta = 0.025/yr):

(1 - 0.025)^{\ \ \ } = _____

7. SA BOL required:

P_{\text{SA,BOL}} = \frac{P_{\text{SA,EOL}}}{(1-\delta)^n} = \frac{\ \ \ \ \ }{\ \ \ \ \ } = _____ W

8. SA area:

A = \frac{P_{\text{BOL}}}{0.295 \times 1361 \times \cos(\ \ \ ) \times 0.85} = \frac{\ \ \ \ \ }{\ \ \ \ \ } = _____ m^2

9. SA type needed (compare to CubeSat reference values):

- [ ] Body-mounted only (1U: ~2 W, 3U: ~7 W, 6U: ~12 W)
- [ ] Single deployable (1U: ~4 W, 3U: ~15 W, 6U: ~30 W)
- [ ] Dual deployable (3U: ~25 W, 6U: ~48 W)

10. SA mass:

m_{\text{SA}} = A \times \sigma = _____ \times _____ = _____ kg

(body-mounted: \sigma = 2.5 kg/m^2; deployable: \sigma = 1.5 kg/m^2)

Part B: Battery Sizing (10 min)

Show all calculations:

1. Eclipse energy:

E = P_{\text{ecl}} \times \frac{t_{\text{ecl}}}{60} = _____ \times _____ = _____ Wh

2. Battery capacity required:

C = \frac{E}{DOD \times \eta} = \frac{\ \ \ \ \ }{0.30 \times 0.95} = \frac{\ \ \ \ \ }{0.285} = _____ Wh

3. With 20% margin: C_{spec} = _____ $\times$ 1.2 = _____ Wh

4. Battery mass:

$m_{\text{bat}} = \frac{C_{\text{spec}}}{E_{\text{specific}}} = \frac{\ \ \ \ }{150} =$ _____ kg

5. Cycle life check:

Number of eclipses in mission = _____ orbits/day $\times$ 365 $\times$ _____ years = _____ cycles

At DOD = 30%, Li-ion provides ~10,000 cycles. Is this sufficient? Y / N

Part C: Thermal Analysis (15 min)

1. Identify hot case and cold case for your orbit:

Parameter	Hot Case	Cold Case
Solar illumination		
Operational mode		
Internal dissipation		
Coating condition	BOL / EOL	BOL / EOL
Key concern		

2. From SpaceCDF Dashboard, record thermal predictions:

Component	Min Predicted (degC)	Max Predicted (degC)	Operating Min (degC)	Operating Max (degC)
Payload / sensor				
Battery				
OBC				
Transponder				

3. Margin check (need >= 5 degC):

4. If any margin fails, what thermal control action would you take?

Part D: SpaceCDF Verification (15 min)

Compare your hand calculations to SpaceCDF values:

Component	Hot margin = Max_op - Max_pr	Cold margin = Min_pred - Min_	Pass?
Parameter	Hand Calculation	SpaceCDF Value	Difference
SA power required (BOL)	_____ W	_____ W	_____ W
Battery capacity	_____ Wh	_____ Wh	_____ Wh
Orbit-average power	_____ W	_____ W	_____ W
Eclipse duration	_____ min	_____ min	_____ min

If there are significant differences, explain why:

____

____

Notes & Reflections

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Worksheet 3.2: Attitude and Orbit Control System (AOCS)

Name:	____

Date:	____
Team / Mission:	____
Role / Position:	____

SpaceCDF Tabs: Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

Key Equations Reference

> *Gravity gradient torque:* $T_{gg} = \frac{3\mu}{2a^3}|I_z - I_x|\sin(2\theta)$
>
> *Aerodynamic torque:* $T_{\text{aero}} = \frac{1}{2}\rho v^2 C_D A \cdot d_{cp-cm}$
>
> *SRP torque:* $T_{\text{SRP}} = \frac{S}{c} A_s (1+q) \cdot d_{sp-cm}$
>
> *Magnetic torque:* $T_{\text{mag}} = M \times B$
>
> *Momentum storage:* $H = T_{\text{dist}} \times t_{\text{desat}}/2$
>
> *Pointing error (RSS):* $\theta_{\text{total}} = \sqrt{\sum \theta_i^2}$
>
> *MTQ torque:* $T_{\text{MTQ}} = m_{\text{dipole}} \times B \times \sin(\alpha)$

Part A: AOCS Architecture Selection (10 min)

Pointing requirement from your mission: ______ deg (3-sigma)

AOCS architecture selected (tick one):

- [ ] Passive magnetic (> 5 deg)
- [ ] Magnetorquers only (2--5 deg)
- [ ] Reaction wheels + MTQ (0.1--2 deg)
- [ ] Fine pointing: RW + star tracker + MTQ (< 0.1 deg)
- [ ] Very fine: RW + ST + gyro + MTQ (< 0.01 deg)

Justification for selection:

____

Estimated AOCS mass: _____ kg   Estimated AOCS power: _____ W

Part B: Disturbance Torque Calculations (20 min)

Calculate at least 2 disturbance torques for your spacecraft and orbit. Show all working.

Spacecraft properties:

Parameter	Value	Unit
Mass		kg
Form factor		U
I_z (largest MOI)		kg m^2
I_x (smallest MOI)		kg m^2
Cross-section area (largest face)		m^2
d_{cp-cm} (CP-CM offset estimate)		m
Orbit altitude		km

1. Gravity gradient torque:

$T_{gg} = \frac{3 \times 3.986 \times 10^{14}}{2 \times (\ \ \ \ )^3} \times |I_z - I_x| =$

$T_{gg} =$ _____ N m

2. Aerodynamic torque (ρ at _____ km $\approx$ _____ kg/m^3):

$T_{\text{aero}} = \frac{1}{2} \times$ _____ $\times$ _____ $^2 \times 2.2 \times$ _____ $\times$ _____ $=$

$T_{\text{aero}} =$ _____ N m

3. (Optional) Solar radiation pressure torque:

$T_{\text{SRP}} =$ _____ N m

4. (Optional) Residual magnetic dipole torque ($M \approx$ _____ A m^2, $B \approx 3 \times 10^{-5}$ T):

$T_{\text{mag}} =$ _____ $\times$ _____ $=$ _____ N m

Summary:

Source	Torque (N m)
Gravity gradient	
Aerodynamic	
SRP	
Magnetic dipole	
Total (RSS or sum)	

Which disturbance dominates? ____

Part C: Reaction Wheel Sizing (10 min)

1. Torque requirement (\geq 2\times total disturbance):

T_{\text{RW,req}} = 2 \times _____ = _____ N m = _____ mN m

2. Momentum storage (desaturation interval = 1 orbit = _____ s):

H = T_{\text{dist}} \times \frac{t_{\text{desat}}}{2} = _____ \times \frac{\ \ \ \ }{2} = _____ mN m s

3. Selected reaction wheel:

Parameter	Required	Selected Product
Torque	_____ mN m	_____ mN m
Momentum	_____ mN m s	_____ mN m s
Mass	--	_____ g
Power	--	_____ W
Quantity	4 (3+1 redundancy)	

Part D: Pointing Error Budget (15 min)

Complete the RSS pointing error budget:

Error Source	Value (deg)	Value^2
Sensor accuracy		
Actuator resolution		
Alignment knowledge		
Thermal distortion		
Jitter (RW)		

Orbit knowledge		
Timing error		
Sum of squares		
RSS Total = \sqrt{\text{sum}}	= _____ deg	

Requirement: _____ deg

Margin: _____ deg (_____ %)

Does the pointing budget close? Y / N

Which error source dominates? ____

What action would most improve the budget? _______________________________

____

Part E: SpaceCDF Verification (10 min)

Compare your hand calculations to SpaceCDF's Pointing Budget display:

Parameter	Hand Calculation	SpaceCDF	Match?
RSS pointing error	_____ deg	_____ deg	
Dominant contributor	____________	____________	
Margin to requirement	_____ %	_____ %	

If there are differences, explain:

____

____

Notes & Reflections

____

____

____

____

____

Worksheet 3.3: Communications and Link Budget Design

Name:	______________________________
Date:	______________________________
Team / Mission:	______________________________
Role / Position:	______________________________

SpaceCDF Tabs: Link Budget, Spectrum Selector, Equipment Browser (Comms)

Key Equations Reference

> *EIRP: \text{EIRP} = P_{TX} + G_{TX} - L_{TX}   (dBW)*
>
> *FSPL: \text{FSPL} = 20\log_{10}\left(\frac{4\pi d f}{c}\right)   (dB)*
>
> *G/T: G/T = G_{RX} - 10\log_{10}(T_{sys})   (dB/K)*
>
> *C/N_0: C/N_0 = \text{EIRP} - \text{FSPL} - L_{\text{losses}} + G/T + 228.6   (dBHz)*
>
> *E_b/N_0: E_b/N_0 = C/N_0 - 10\log_{10}(R_b)   (dB)*
>
> *Link margin: \text{Margin} = E_b/N_{0,\text{avail}} - E_b/N_{0,\text{req}} - L_{\text{impl}} \geq 3 dB*
>
> *Antenna gain: G = \eta_a(\pi D/\lambda)^2*
>
> *dB conversion: P_{\text{dBW}} = 10\log_{10}(P_W);   2 W = +3.0 dBW; 1 W = 0 dBW*

Part A: Frequency Band Selection (10 min)

Mission data rate requirement: _____ Mbps

License type selected: Amateur / Experimental / Commercial

Band selected: _____________   Centre frequency: _____ MHz

Rationale for band selection:

____

____

Licensing/filing required: ____

Estimated licensing cost: _____   Timeline: _____ months

Part B: Complete Link Budget (25 min)

Fill in ALL rows. Show computation for EIRP, FSPL, G/T, C/N_0, and margin.

Line	Parameter	Formula / Source	Value	Unit
1	TX Power	_____ W =		dBW
2	TX Antenna Gain			dBi
3	TX Line Losses			dB
4	EIRP = Line 1 + 2 + 3		______	dBW
5	Free Space Path Loss	20\log_{10}(4\pi \times __		dB
6	Atmospheric Loss			dB
7	Pointing Loss			dB
8	Polarisation Loss			dB
9	RX Antenna Gain			dBi
10	System Noise Temp	_____ K =		dBK
11	G/T = Line 9 - 10		______	dB/K
12	Boltzmann Constant		+228.6	dBW/K/Hz
13	C/N_0 = 4 + 5 + 6 + 7 + 8		______	dBHz
14	Data Rate	_____ bps = 10\log_{10}(		dBbps
15	E_b/N_0 available = 13 - 1		______	dB
16	E_b/N_0 required (mod+c			dB
17	Implementation Loss			dB
18	LINK MARGIN = 15 - 16 -		______	dB

Show FSPL calculation:

\text{FSPL} = 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c)

$= 21.98 + 20\log_{10}(\_\_\_\_\_) + 20\log_{10}(\_\_\_\_\_) - 169.54 =$ _____ dB

Does the link close (≥ 3 dB)? Y / N

If margin is excessive, maximum achievable data rate at 3 dB margin:

$R_{b,\text{max}} = 10^{(E_b/N_{0,\text{avail}} - E_b/N_{0,\text{req}} - L_{\text{impl}} - 3)/10} \times R_b =$ _____ Mbps

---

Part C: Data Budget (10 min)

Daily data generation:

$V_{\text{gen}} = R_{\text{payload}} \times t_{\text{imaging}} \times N_{\text{orbits}} \times f_{\text{compression}}$

= _____ Mbps $\times$ _____ s $\times$ _____ $\times$ _____ = _____ MB/day = _____ GB/day

Daily downlink capacity:

$V_{\text{DL}} = R_{\text{DL}} \times t_{\text{contact}} \times N_{\text{passes}} \times \eta_{\text{protocol}}$

= _____ Mbps $\times$ _____ s $\times$ _____ $\times$ _____ = _____ MB/day = _____ GB/day

Data budget closes? ($V_{\text{DL}} \geq V_{\text{gen}}$): Y / N

If no, what is your mitigation plan?

Part D: SpaceCDF Comparison (10 min)

Open the Link Budget tab, enter your parameters, and compare:

Parameter	Hand Calculation	SpaceCDF	Difference
EIRP (dBW)			

FSPL (dB)			
C/N_0 (dBHz)			
Link Margin (dB)			

If there are differences, explain:

____

____

Part E: Modulation and Coding Selection (5 min)

Selected modulation: ____

Selected FEC code: ____

Required E_b/N_0 at BER 10^{-6}: _____ dB

Spectral efficiency: _____ bps/Hz

Rationale:

____

____

Notes & Reflections

____

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____

Worksheet 3.4: Structure, Propulsion, and Equipment Selection

Name:	____
Date:	____
Team / Mission:	____
Role / Position:	____

SpaceCDF Tabs: Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

---

Key Equations Reference

> *Structural Margin of Safety:* $\text{MoS} = \frac{\sigma_{\text{allow}}}{\sigma_{\text{design}} \times \text{FoS}} - 1 \geq 0$
>
> *   FoS: Yield 1.25, Ultimate 1.5 (metallic)*
>
> *Tsiolkovsky rocket equation:* $\Delta V = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$
>
> *Propellant mass:* $m_{\text{prop}} = m_{\text{dry}} \times \left(e^{\Delta V/(I_{sp} \cdot g_0)} - 1\right)$
>
> * * $g_0 = 9.80665\ m/s^2$
>
> *Data storage:* $S = V_{\text{daily}} \times N_{\text{days}} \times f_{\text{safety}}$
>
> *Fundamental frequency:* $f_1 > 40$ *Hz (deployer requirement)*

Part A: CDS Compliance Check (10 min)

CDS Requirement	Your Design Value	Specification	Compliant?
Form factor: _____ U	Dimensions: _____ mm	_____ mm	Y / N
Maximum mass	_____ kg (wet)	_____ kg	Y / N
Rail material		Hard anodised Al	Y / N
Deployment switches	_____ (quantity)	Min 2	Y / N
RBF pin provided	Y / N	Required	Y / N
No protrusions beyond rails (stov	Y / N	Required	Y / N

CG within 2 cm of geometric cen	Estimated: _____ cm offset	<= 2 cm	Y / N
Fundamental frequency	Estimated: _____ Hz	> 40 Hz	Y / N

Non-compliance items and corrective actions:

____

____

Part B: Structural Margin Calculation (10 min)

Axial launch load: _____ g   Lateral launch load: _____ g

Load per rail (4 rails, equal sharing):

F_{\text{axial}} = \frac{m \times a_{\text{axial}} \times g_0}{4} = \frac{\ \ \ \times \ \ \ \ \times 9.81}{4} = _____ N

Stress in rail (cross-section 8.5 x 8.5 mm = 7.225 \times 10^{-5} m^2):

\sigma = \frac{F}{A} = \frac{\ \ \ }{7.225 \times 10^{-5}} = _____ MPa

Margin of safety (yield, Al 7075-T6, \sigma_y = 503 MPa):

\text{MoS} = \frac{503}{\ \ \ \times 1.25} - 1 = _____

Pass? Y / N

____

Part C: Propulsion Decision (15 min)

Does your mission need propulsion?

Question	Answer
Orbit altitude	_____ km
Natural orbital lifetime	_____ years
FCC 5-year compliant without propulsion?	Y / N
Orbit maintenance required?	Y / N
Constellation phasing required?	Y / N
Active deorbit required?	Y / N

Decision: Propulsion / No propulsion

If propulsion is selected, complete this section. Show all working.

Total ΔV required:

Manoeuvre	ΔV (m/s)
Orbit maintenance (_____ yr)	
Deorbit	
Collision avoidance margin	
Total	

Propulsion type selected: ________________ I_{sp} = _____ s

Propellant mass (Tsiolkovsky):

$m_{\text{prop}} = m_{\text{dry}} \times \left(e^{\Delta V/(I_{sp} \times g_0)} - 1\right)$

= _____ $\times \left(e^{\ \ \ /(\ \ \ \times 9.81)} - 1\right)$ = _____ $\times \left(e^{\ \ \ } - 1\right)$ = _____ $\times$ _____ = _____ kg

____

____

System dry mass (thruster + tank + feed): _____ kg

Total propulsion system mass: _____ kg (propellant + dry)

Fraction of total spacecraft mass: _____ %

Part D: Equipment Selection Log (25 min)

Using SpaceCDF's Equipment Browser, select components for each category:

Category	Component Se	Manufacturer	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Total Mass
EPS Board						1	
Battery						1	
Solar Panels							
OBC						1	
Reaction Whee						x	
Magnetorquer						x	
Star Tracker							
Sun Sensor						x	
Transponder						1	
Antenna							
Payload						1	
Structure						1	

Harness						1	
Propulsion							
TOTAL			_____		_____		

Parametric estimate (from Session 2.4): _____ kg

Equipment-based total: _____ kg

Difference: _____ kg (_____ %)

Any incompatibilities detected? (RF band mismatch, voltage mismatch, interface conflict)

____

____

Part E: Component Trade Study (15 min)

Subsystem traded: _____________   Options compared: 3

Criterion	Weight	Option A: ___	Score	Option B: ___	Score	Option C: ___	Score
Mass	0.20						
Power	0.15						
Cost	0.20						
TRL	0.15						
Heritage	0.15						
Performance	0.15						
Weighted	1.00		_____		_____		_____

Winner: _____________   Score: _____

Rationale (beyond the numbers):

____

____

____

Part F: Final Budget Health Check (10 min)

After equipment selection, record final budget status:

Budget	Value	Margin	Status
Mass (wet vs allocation)	_____ / _____ kg	_____ %	G / A / R
Power (worst mode vs SA)	_____ / _____ W	_____ %	G / A / R
Link margin	_____ dB	>= 3 dB?	G / A / R
Cost vs ceiling	_____ / _____ MEUR	_____ %	G / A / R
Pointing (RSS vs req)	_____ / _____ deg	_____ %	G / A / R
Data (DL vs gen)	_____ / _____ GB/day	_____ %	G / A / R

Any budget that does not close? ____

Proposed fix: ____

____

Notes & Reflections

____

____

____

____

____

____

____

____

Worksheet 4.1: Equipment Selection & Bill of Materials

Name: __________________________ Date: __________ Team: __________

Mission Name: __________________________ Form Factor: ______U

Part A: Make/Buy/Reuse Decisions

For each subsystem, document the procurement decision and rationale:

Subsystem	Decision (Buy/Reuse/Make)	Rationale	Lead Time
Structure			
EPS (solar array)			
EPS (battery)			
EPS (distribution)			
OBC			
AOCS (sensors)			
AOCS (actuators)			
TTC (transceiver)			
TTC (antenna)			
Thermal			
Payload			
Propulsion (if applicable)			

Part B: Bill of Materials (BOM)

Complete the BOM from the SpaceCDF Equipment Browser selections:

Part C: Budget Tracking

After all selections, record the live budget values from the SpaceCDF Dashboard:

Item ID	Component	Manufactur	Part Numbe	Qty	Mass (g)	Power (W)	Cost (kEUR	TRL	ECCN
STR-001									
EPS-001									
EPS-002									
EPS-003									
OBC-001									
AOCS-001									
AOCS-002									
AOCS-003									
TTC-001									
TTC-002									
THM-001									
PLD-001									

HRN-001									
Budget	BOM Total	Parametric E	Allocation	Margin	Margin %				
Dry mass	_____ g	_____ g	_____ g	_____ g	_____%				
Peak power	_____ W	_____ W	_____ W	_____ W	_____%				
Total cost	_____ kEUF	_____ kEUF	_____ kEUF	_____ kEUF	_____%				
Volume	_____% of _		100%	_____%					

Does everything fit within allocation? Y / N

If not, what would you change? ____

____

Part D: Export Control Assessment

List any components that may require export licences:

Component	ECCN / USML Category	Controlled?	Licence Required?	Action Needed
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	

Are there any ITAR-controlled items? Y / N

If yes, what TAA (Technical Assistance Agreement) or DSP-5 licence is needed?

____

Part E: Interface Compatibility Check

For each selected component, verify interface compatibility:

Component	Bus Voltage OK?	Data Protocol OK?	RF Band Match?	Connector OK?	Notes
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	

Part F: Interface Incompatibility Resolution

Describe one interface incompatibility found during selection and how you resolved it:

Issue: ____

____

Components involved: ____

Root cause: ____

Resolution chosen: ____

Impact on budget: ____

Notes & Reflections

What was the most difficult equipment selection decision? Why?

____

____

____

____

____

____

____

____

Worksheet 4.2: Verification & Validation Planning

Name: __________________________ Date: __________ Team: __________

Mission Name: __________________________

Part A: IADT Method Assignment

For 10 key requirements from your SpaceCDF V&V Matrix, assign the verification method and justify:

#	Req ID	Requirement Tex	Method (I/A/D/T)	Phase (B/C/D)	Level (Unit/Sub/S	Justification
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Summary count: Analysis: ___ / Test: ___ / Demonstration: ___ / Inspection: ___

Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis (early confidence) AND test (final proof):

Part C: Environmental Test Sequence

For your selected launch vehicle, specify the complete proto-flight test sequence:

Launch vehicle: _______________ PUG reference: _______________

Requirement ID	Requirement Text	Analysis (Phase B)	Test (Phase C/D)	Why Both Needed?
Step	Test	Specification	Duration	Pass Criteria
1	Initial functional test	Full performance baseline	_____ hr	All parameters nominal
2	Sine vibration	_____ g, _____ - _____ H	_____ min/axis x 3	No resonance shift > 5%
3	Random vibration	_____ gRMS overall	_____ min/axis x 3	No structural damage
4	Post-vibe functional test	Same as step 1	_____ hr	Compare to baseline
5	Thermal vacuum -- hot	+ _____ C (predicted max	_____ cycles	Functional at hot extreme
6	Thermal vacuum -- cold	- _____ C (predicted min	_____ cycles	Functional at cold extreme
7	Post-TVAC functional test	Same as step 1	_____ hr	Compare to baseline
8	EMC test	Required? Y / N	_____ day	No interference
9	Mass measurement	Calibrated scale (+/- 0.01	30 min	M <= _____ kg

10	Deployer fit check	_____ deployer	2 hr	Fits; slides freely

Total test campaign duration estimate: _____ weeks

Part D: Vibration Test Profile

Record the random vibration PSD profile from your launch vehicle PUG:

Frequency (Hz)	PSD Level (g^2/Hz)	Slope
20		Start
20 - _____		+ _____ dB/oct
_____ - _____		Flat
_____ - 2000		- _____ dB/oct
2000		End

Overall gRMS (calculated): _____ gRMS

Show your calculation:

____

____

____

Part E: TVAC Cycle Profile

Sketch or describe one TVAC cycle:

Hot extreme: + _____ C   Cold extreme: - _____ C

Ramp rate: _____ C/min   Dwell time at extreme: _____ hr

Functional test at each extreme? Y / N   Duration of functional test: _____ hr

Calculate time for one cycle:

t_cycle = ____

= _____ hours

Total TVAC campaign: _____ cycles x _____ hr + _____ hr setup = _____ hr = _____ days

Part F: Waiver Identification

Are there any requirements that cannot be verified by the standard method? Document potential waivers:

Notes & Reflections

Which environmental test do you consider most critical for your mission? Why?

____

____

____

____

____

____

____

____

Worksheet 4.3: Risk, Interfaces & FMECA

Name: ______________________________ Date: ____________ Team: ____________

Mission Name: ______________________________

Part A: Risk Register

Identify at least 5 technical risks for your mission design. Score each on the 5x5 matrix.

Scoring reminder: L x C = Score. Low (1-4), Medium (5-9), High (10-15), Critical (16-25).

Req ID	Standard Me	Proposed Alt	Justification	Risk				

#	Risk Descripti	L (1-5)	C (1-5)	Score	Category (L/M	Mitigation Stra	Owner (Positi	Post-Mitigation Score
1								
2								
3								
4								
5								
6								
7								

How many risks are High or Critical (before mitigation)? _____

How many remain High or Critical after mitigation? _____

Part B: 5x5 Risk Matrix Plot

Plot your risks on the matrix below. Write the risk number in the appropriate cell.

```
               Consequence ->
               1       2       3       4       5
    L   5 |       |       |       |       |       |
    i   4 |       |       |       |       |       |
    k   3 |       |       |       |       |       |
    e   2 |       |       |       |       |       |
    l   1 |       |       |       |       |       |
```

Part C: N-Squared (N^2) Interface Matrix

Fill in the interface matrix for your mission's subsystems. Use abbreviations:

- P = Power (electrical)
- D = Data (I2C, SPI, UART, CAN, RS-422)
- R = RF (coaxial)
- M = Mechanical (mounting, alignment)
- T = Thermal (conductive/radiative)

	EPS	OBC	AOCS	TTC	Payload	Structure
EPS	---					
OBC		---				
AOCS			---			
TTC				---		
Payload					---	
Structure						---

Total number of interfaces identified: _____

Interface conflicts detected: (describe at least 2)

1. Conflict: ____

Resolution: ____

2. Conflict: ____

Resolution: ____

Part D: FMECA Table

Complete for 4 critical components:

Part E: Single-Point Failure Analysis

List all single-point failures in your design:

Component	Function	Failure Mode	Cause	Local Effect	System Effec	Severity (1-5	Detection Me	Compensatir
#	Component	Failure Mode	Effect on Miss	SPF?	Acceptable?	Mitigation (if r		
1				Y/N	Y/N			
2				Y/N	Y/N			
3				Y/N	Y/N			
4				Y/N	Y/N			
5				Y/N	Y/N			

Total SPFs identified: _____ Accepted: _____ Mitigated: _____

Part F: Reliability Calculation

Calculate the series reliability for your mission's critical chain (components that must all work for mission success):

R_series = R_1 x R_2 x ... x R_n = _____________________________

R_series = ___________

If you added redundancy to one component (e.g., dual deployment mechanism), recalculate:

R_redundant = 1 - (1 - R)^2 = 1 - (1 - _____)^2 = _____

New R_series (with redundancy) = _____

Improvement: from _____ to _____ (+ _____ percentage points)

Part G: Risk Discussion

What is the highest-risk item in your design? What would it take (cost, mass, schedule) to bring the risk to an acceptable level?

____

____

____

____

____

____

Notes & Reflections

Which single-point failure concerns you the most? Would you accept it, or is mitigation essential?

____

____

____

____

____

____

____

____

Worksheet 4.4: Cost Estimation & Design Review

Name: ____________________________ Date: ___________ Team: ___________

Mission Name: ____________________________

Part A: Mission Cost Estimate (WBS)

Complete using both parametric estimates (from SpaceCDF Cost tab) and bottom-up estimates (from BOM vendor quotes):

Component	Individual Reliability	Mission Duration Us			
WBS #	Element	Parametric (kEUR)	Bottom-Up (kEUR)	Difference	Notes
1.0	Programme Manager				
2.0	Systems Engineering				
3.0	Mission Assurance				
4.0	Payload				
5.0	Spacecraft Bus Hardv				
	5.1 Structure & mech				
	5.2 EPS				
	5.3 AOCS				
	5.4 TTC				
	5.5 OBC & data hand				
	5.6 Thermal				
	5.7 Propulsion				
	5.8 Harness				
6.0	Integration & Test				
7.0	Software (FSW + GS				
8.0	Launch Services				
9.0	Ground Segment				
10.0	Operations (_____ ye				
	TOTAL	_____	_____		

Part B: Confidence Levels

P50 estimate (your best estimate): _____ kEUR

P70 estimate (P50 x 1.2): _____ kEUR

P80 estimate (P50 x 1.3): _____ kEUR

Show your reasoning for the multiplier (is 1.3 appropriate, or should it be higher for your mission?):

____

____

____

Part C: Constellation Cost (if applicable)

Number of satellites: _____ Learning rate: _____%

b = ln(learning_rate) / ln(2) = ln(_____) / ln(2) = _____

Unit #	Unit Cost (kEUR) = C_1 x N^b	Cumulative (kEUR)
1		
2		
4		
8		
_____ (total)		

Show your calculation for the total constellation cost:

Hardware: _____ units x _____ kEUR avg = _____ kEUR

Spares (____%): _____ units x _____ kEUR = _____ kEUR

Launch: _____ sats x _____ kEUR = _____ kEUR

Ground segment: _____ kEUR

Operations (_____ years): _____ kEUR

PM/SE/MA: _____ kEUR

Total constellation cost: _____ kEUR = _____ MEUR

P80 constellation cost (x 1.3): _____ MEUR

Part D: Cost Drivers

Top 3 cost drivers in your mission (from SpaceCDF Cost Breakdown):

Rank	Cost Element	Cost (kEUR)	% of Total	Why is it Expensive?
1				
2				
3				

What would you change to reduce total cost by 20%?

____

____

____

____

Part E: Design Review Readiness

From the SpaceCDF Gate Review tab, assess your readiness for SRR/PDR:

#	Gate Criterion	Status (Pass/Fail/Manua	Evidence Location	If Fail: Action Needed
1	Mission need justified			
2	Requirements complete &			
3	All budgets close with ma			
4	Equipment selected (BOM			
5	Interfaces defined, conflic			
6	V&V approach defined			
7	Risks identified with mitiga			
8	Cost estimate with WBS			
9	ConOps defined			
10	Sustainability compliance			

Must-pass criteria met: _____ / _____

Overall readiness: READY / NOT READY

Action items to achieve readiness:

1. ____
2. ____
3. ____

Part F: Presentation Preparation

Outline your 12-minute design review presentation:

Section	Duration	Key Message	Evidence/Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Notes & Reflections

What is the most uncertain element in your cost estimate? How would you reduce the uncertainty?

____

____

____

____

____

____

____

____

Worksheet 5.1: Ground Segment & Operations Architecture

Name: ______________________________ Date: ____________ Team: ____________

Mission Name: ______________________________ Orbit: _____ km, _____ deg inclination

---

Part A: Ground Station Contact Analysis

Calculate contact time and data volume per pass for your mission.

Input Parameters

Parameter	Value	Source
Orbital altitude (h)	_____ km	SpaceCDF orbit selection
Minimum elevation angle (epsilon)	_____ deg	Ground station spec (typically 5-10)
Downlink data rate (R)	_____ Mbps	SpaceCDF link budget
Protocol efficiency (eta)	_____ (typically 0.85-0.95)	CCSDS overhead estimate
Ground station latitude	_____ deg	Selected station
Earth radius (R_E)	6371 km	Constant

Calculations

Maximum slant range:

rho = R_E x (sqrt((h/R_E + 1)^2 - cos^2(epsilon)) - sin(epsilon))

rho = 6371 x (sqrt((_____/6371 + 1)^2 - cos^2(_____)) - sin(_____))

rho = _____ km

Average pass duration (approximate):

T_avg ~ _____ minutes (from STK/SpaceCDF or estimate: ~6 min for 500 km SSO)

Data volume per pass:

V_data = R x T x eta = _____ Mbps x _____ s x _____ = _____ Mbit = _____ MB

Part B: Data Budget Closure

Parameter	Value	Unit
Daily data generation (payload)		MB/day
Daily data generation (housekeeping)		MB/day
Total daily generation		MB/day
Passes per day (GS 1: _____________)		passes
Data per pass (GS 1)		MB
Subtotal downlink (GS 1)		MB/day
Passes per day (GS 2: _____________)		passes
Data per pass (GS 2)		MB

Subtotal downlink (GS 2)		MB/day
Total daily downlink capacity		MB/day
Data budget margin		MB/day
Margin percentage		%

Does the data budget close? Y / N

If not, what changes would close it? (check all that apply)

- [ ] Add another ground station at: ________________
- [ ] Increase data rate to: _____ Mbps (requires: ________________)
- [ ] Reduce payload data generation to: _____ MB/day (impact: ________________)
- [ ] Add onboard compression (ratio: _____:1)
- [ ] Use SatNOGS network for additional passes
- [ ] Other: ____

Part C: Ground Station Network Design

Station #	Location	Latitude	Antenna	Band	Data Rate	Passes/Day	Role
1							Primary TTC
2							Payload DL
3							Backup/SatNOGS

Total ground segment cost estimate: _____ kEUR

Cost Element	Annual Cost (kEUR)	Notes
Antenna rental / ownership		
MCS software licence		
Network connectivity		
Operations staff (_____ FTE)		
Total annual		
Mission lifetime (_____ yr)		

Part D: LEOP Timeline

Construct the LEOP timeline for the first 72 hours after separation:

Time (UTC)	Sim Time	Activity	Success Criterion	Status
T+0	Separation	Deployment switches rele		[ ]
T+30 min	Timer expires	Antenna deployment com		[ ]

T+_____		Beacon acquisition	Carrier lock	[ ]
T+_____		First HK telemetry	Data decoded	[ ]
T+_____		First uplink command	ACK received	[ ]
T+_____		SA deployment (if separat	Power generation confirm	[ ]
T+_____		ADCS initialisation	Attitude determination act	[ ]
T+_____		ADCS calibration	Pointing < _____ deg	[ ]
T+_____		Full duplex validation	Uplink + downlink at ops r	[ ]
T+_____		Health assessment compl	All subsystems nominal	[ ]

Part E: Mission Operations Timeline (Gantt Chart)

Sketch the operations timeline from launch to end of life:

```
Phase:       LEOP  Commission  Early Ops    Nominal Operations         EOL
Week:     1  2  3  4  5  6  7  8  9  ... 26 ... 52 ... 104 ... 156
          |  |  |  |  |  |  |  |  |       |      |       |       |
Staffing: ________________________________________________
          24/7     16/7        8/5         8/5 or automated    8/5

Key milestones:
  L+___: First light
  L+___: Commissioning complete
  L+___: Orbit maintenance manoeuvre (if applicable)
  L+___: End of nominal mission
  L+___: Extended mission (if applicable)
  L+___: Passivation and deorbit
```

Notes & Reflections

What is the most challenging aspect of your ground segment design? What would you do differently with more budget?

____

____

____

____

____

____

Worksheet 5.2: Mission Operations Concepts

Name: ____________________________ Date: ___________ Team: ___________

Mission Name: ____________________________

Part A: Operational Modes

Define all operational modes for your spacecraft. Use the SpaceCDF ConOps Editor to enter these.

Verify: Does the power budget close in every mode?

Mode	Description	Power (W)	Data Rate	Entry Condition	Exit Condition
Safe					
Detumble					
Standby					
Science					
Downlink					
Eclipse					
Manoeuvre					
Mode	Power Consumption (	Power Available (W)	Margin (W)	Closes?	
Safe				Y / N	
Detumble				Y / N	
Standby				Y / N	
Science				Y / N	
Downlink				Y / N	
Eclipse (battery only)				Y / N	

Part B: FDIR Rules

Define at least 5 FDIR rules for your spacecraft:

#	Fault Descriptior	Detection Methoc	Threshold	FDIR Level (0-4)	Autonomous Res	Recovery Proced
1						
2						
3						
4						
5						
6						

Can Safe Mode sustain the spacecraft indefinitely (power + thermal)? Y / N

If not, what is the maximum Safe Mode duration? _____ hours

Can every autonomous FDIR action be overridden from ground? Y / N

If not, which cannot? ____

Part C: Nominal Procedure

Write one nominal procedure for your mission (science observation or data downlink):

Procedure ID: ___________ Title: ____

Preconditions:

____

____

Step	Action	Expected Result	If Not Achieved
1			
2			
3			
4			
5			
6			
7			
8			

Post-conditions:

____

Part D: Contingency Procedure

Write one contingency procedure (e.g., safe mode recovery, communication loss recovery):

Procedure ID: ____________ Title: ____

Trigger condition:

____

Severity level: Green / Yellow / Orange / Red

Step	Action	Expected Result	If Not Achieved	Timeout
1				
2				
3				
4				
5				
6				

Escalation: If this procedure fails, escalate to: ____

Post-conditions:

____

Part E: Operations Staffing

Estimate the staffing requirements for each mission phase:

Phase	Duration	Shifts/Day	Staff/Shift	Total FTE	Annual Cost (kEUR)
LEOP	_____ days	3 (24/7)			
Commissioning	_____ weeks	2 (16/7)			
Early Operations	_____ months	1 (8/5)			
Nominal Operations	_____ years	1 (8/5) or auto			
End of Life	_____ weeks	1 (8/5)			
Total					

Part F: Anomaly Scenario

Hypothetical scenario: Your satellite enters Safe Mode during a weekend. The next ground contact is

Monday morning (40 hours away). Walk through your response:

1. What is the spacecraft doing in Safe Mode?

2. Will it survive 40 hours? (Check power and thermal)

3. What telemetry will you review first on Monday?

4. What is your recovery plan?

Notes & Reflections

How much autonomy should your spacecraft have? Where is the line between onboard decision-making and ground control?

Worksheet 5.3: Mission Simulation Day

Name: ____________________________ Date: ___________ Team: ___________

Mission Name: ____________________________

Your Role: ____________________________ (FD / SC / GSO / PO / FD-nav / Logger)

Simulation Event Log

Record ALL events, decisions, and actions in real-time during the simulation:

#	Sim Time	Real Time	Event / Observat	Action Taken	Result	Logged By
1	T+0		Separation			
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

LEOP Checklist

Check off each LEOP milestone as it is achieved during the simulation:

#	Milestone	Time Achieved	Status	Notes

1	Separation confirmed		[ ]	
2	30-min timer expired		[ ]	
3	Antenna deployed		[ ]	
4	Beacon acquired		[ ]	
5	First HK telemetry decode		[ ]	
6	First uplink command ACI		[ ]	
7	Solar array deployed (if se		[ ]	
8	ADCS detumble complete		[ ]	
9	ADCS calibration complet		[ ]	
10	Full duplex validated		[ ]	
11	All subsystems nominal		[ ]	

Anomaly Response Form 1

Complete this form for each injected anomaly during the simulation.

Field	Entry
Anomaly ID	ANO-SIM-001
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

____

____

____

Initial Assessment (what do you think happened?):

____

____

____

Diagnosis (root cause after investigation):

____

____

____

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

____

____

Follow-up Required:

____

Procedure used: _______________ Did it work as written? Y / N / Partially

If not, what changes are needed? ____

Anomaly Response Form 2

Field	Entry
Anomaly ID	ANO-SIM-002
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

____

____

____

Initial Assessment (what do you think happened?):

____

____

____

Diagnosis (root cause after investigation):

____

____

____

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

____

____

Follow-up Required:

____

Procedure used: _______________ Did it work as written? Y / N / Partially

If not, what changes are needed? ____

Anomaly Response Form 3

Field	Entry
Anomaly ID	ANO-SIM-003
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

____

____

____

Initial Assessment:

____

____

Diagnosis:

____

____

Action Taken:

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				

Final Result:

____

Procedure used: _______________ Did it work as written? Y / N / Partially

---

Commissioning Status

After the commissioning phase of the simulation, record subsystem checkout status:

Subsystem	Checkout Complete?	Performance vs Spec	Issues Found
EPS	[ ]	Nominal / Degraded / Failed	
OBC	[ ]	Nominal / Degraded / Failed	
AOCS	[ ]	Nominal / Degraded / Failed	
TTC	[ ]	Nominal / Degraded / Failed	
Thermal	[ ]	Nominal / Degraded / Failed	
Payload	[ ]	Nominal / Degraded / Failed	
Propulsion	[ ]	Nominal / Degraded / Failed / NA	

Nominal Operations Summary

After the nominal operations phase, record:

Metric	Planned	Actual (Simulated)	Variance
Science observations completed			
Data downlinked (MB)			
Anomalies encountered			
Ground contacts made			
Safe mode entries			

Simulation Debrief

What went well?

____

____

____

What could be improved?

____

____

____

Design weaknesses revealed by simulation:

____

____

____

Procedure changes needed:

____

____

____

FDIR gaps identified:

____

____

Notes & Reflections

What was the most stressful moment during the simulation? What did it teach you about operations?

____

____

____

____

____

____

____

____

Worksheet 5.4: Final Review & Presentations

Name: ______________________________ Date: ____________ Team: ____________

Mission Name: ______________________________

Part A: Final Design Summary

Record the final state of your design from the SpaceCDF Dashboard:

Parameter	Value	Unit	Margin	Margin %	Status (G/A/R)
Dry mass		kg			
Wet mass (if propulsi		kg			
SA power (EOL)		W			
Battery capacity		Wh			
Link margin (worst ca		dB			
Pointing accuracy		deg			
Data balance		MB/day			
Total cost (P50)		kEUR			
Total cost (P80)		kEUR			
Debris compliance sc		/100			
Mission lifetime		years			

Do all budgets close? Y / N   If not, which fail? ____

Number of unresolved conflicts: _____

Part B: Documentation Checklist

Tick each document generated from SpaceCDF Exports tab:

Document	Generated?	Complete?	Notes
MRD (Mission Requirements Do	[ ]	[ ]	
Technical Specification	[ ]	[ ]	
Verification Plan	[ ]	[ ]	
ConOps Document	[ ]	[ ]	
SEMP (Systems Engineering Mg	[ ]	[ ]	
IRD (Interface Requirements Do	[ ]	[ ]	
Risk Management Plan	[ ]	[ ]	
Test Plan	[ ]	[ ]	
Bill of Materials	[ ]	[ ]	
ITU Filing Template	[ ]	[ ]	
RSSSA Filing	[ ]	[ ]	
Export Control Assessment	[ ]	[ ]	
End-of-Life Report	[ ]	[ ]	
COPUOS Registration	[ ]	[ ]	

Total documents generated: _____ / 14

Part C: Presentation Outline

Prepare your 12-minute final design review presentation:

Section	Time	Key Points to Cover	Evidence / Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Presenter(s): ____

Backup slides prepared on: ____

Part D: Peer Review Evaluation

Complete this section while reviewing another team's presentation.

Team being reviewed: _________________________

Reviewers: _________________________

#	Criterion	Score (0-2)	Comments / Questions
1	Mission need clearly justified		
2	Requirements complete and trac		
3	All budgets close with margin		
4	Equipment selected with justified		
5	Interfaces defined, conflicts reso		
6	V&V approach defined		
7	Risks identified with mitigations		
8	Cost estimate with WBS and P8(		
9	ConOps and ground segment de		
10	Sustainability compliance		
	TOTAL	/ 20	

Presentation quality:

Criterion	Score (1-5)	Comments
Clarity of communication		
Use of evidence (data, not assertions)		
Ability to answer questions		
Time management		

Board recommendation: GO / NO GO / GO WITH ACTIONS

Action items for the presenting team:

#	Action	Responsible	Deadline
1			
2			
3			

Strongest aspect of this team's design:

____

Biggest area for improvement:

____

Part E: Lessons Learned

1. What was the most challenging design decision your team faced?

____

____

____

2. What would you do differently if starting the course over?

____

____

____

3. What does the design need next (Phase B activities)?

____

____

____

4. What was the most valuable SpaceCDF feature?

____

____

5. What feature was missing that you needed?

____

____

6. What concept from this course will you apply in your future work?

____

____

____

Part F: Self-Assessment Rubric

Rate your own learning across the three weeks of the course. Be honest -- this is for your own reflection, not grading.

Competency	Week 1 Start (1-5)	Now (1-5)	Evidence of Growth
Define a space mission starting f			
Write verifiable, traceable require			
Conduct a structured trade study			
Size subsystems using parametr			
Select COTS equipment and ver			
Construct a verification matrix wi			
Assess risk using a 5x5 matrix a			
Estimate mission cost using para			
Design a ground segment and ca			
Define operational modes, FDIR			
Respond to anomalies using a sy			
Present a design to a review boa			

Scale: 1 = No familiarity, 2 = Basic awareness, 3 = Can apply with guidance, 4 = Can apply independently, 5 = Can teach others

Overall self-assessment:

At the start of this course, I would describe my systems engineering ability as:

____

Now, I would describe it as:

____

One thing I am most proud of from this course:

____

Notes & Reflections

Final thoughts on the course, the design process, and what comes next: